

TOPIC: PROGRAMMING LANGUAGES (SS1)

====>A programming language is an artificial language that is capable of controlling the behavior of a machine, particularly a computer system. Just like human languages; programming languages have predefined rules of operations.

====> A programming language is a formal set of instructions that can be used to produce various kinds of output, including computer programs, algorithms, and data processing.

PURPOSES OF USES PROGRAMMING LANGUAGES

- To communicate with computers.
- To create software applications, systems, and scripts that solve problems.

LEVELS OF PROGRAMMING LANGUAGES: The different levels of programming languages are:

1. HIGH-LEVEL LANGUAGES:

These are Languages that are closer to human languages, making them easier to read and write. It is easier to use and more portable across platforms.

Examples include JavaScript, Java, C#, Python, etc.

FEATURES:

- Abstraction from hardware.
- Improved productivity.
- Automatic memory management.

MODES OF EXECUTION:

- (a). Interpreted: They are read and then executed directly with no compilation stage.
- (2). Compiled languages: They are transformed into an executable form before running.
- (3). Translated: High-level languages are translated into low-level languages which the computer understands.

2. LOW-LEVEL LANGUAGES:

These are languages that provide little or no abstraction from a computer's microprocessor and are closer to machine code. These are divided into two parts:

(a). Machine Language/Code: Is a system of instructions and data directly understandable by a computer's central processing unit. It is usually made up of binary codes.

(b). Assembly Language: Is a human-readable representation of machine code. e.g. "ADD" for addition, "SUB" for subtraction, START, LABEL, etc. An assembly language is also called SYMBOLIC PROGRAMMING LANGUAGE. The computer program for translating assembly Language into object code is called ASSEMBLER.

CHARACTERISTICS OF MACHINE LANGUAGE:

- i). It is machine dependent.
- ii). It uses special codes and the assignment of storage devices.

CHARACTERISTICS OF ASSEMBLY LANGUAGE:

- i). Machine dependent.
- ii). Uses symbolic operation codes.

CHARACTERISTICS OF HIGH-LEVEL LANGUAGE:

- i). Machine dependent
- ii). Uses instructions that seem English-like.
- iii). One-to-many Language.

ADVANTAGES OF HIGH-LEVEL LANGUAGE

- a). User-friendly
- b). Easy to learn
- c). Similar to English
- d). Requires less time to write
- e). Easier to maintain.

DISADVANTAGES OF HIGH-LEVEL LANGUAGE

- a). It has to be translated by a translator to machine Language and thus a price in computer time is paid.
- b). The object code generated by a translator might be inefficient compared to an equivalent assembly Language program.

COMPARISONS BETWEEN LOW-LEVEL LANGUAGES AND HIGH-LEVEL LANGUAGES

==> LLL (Compiled Languages): Translated into machine code before execution. (e.g., C, C++). Faster execution but requires compilation.

==> HLL (Interpreted Languages): Translated during runtime. (e.g., Python, JavaScript). Easier debugging but typically slower.

SOME USEFUL TERMS IN PROGRAMMING:

*SYNTAX: The set of rules that defines the combinations of symbols that are considered to be a correctly structured program.

*SEMANTICS: The meaning of the constructs in a programming language.

*IMPERATIVE: Focuses on how to perform tasks (e.g., C, Java).

*DECLARATIVE: Focuses on what the program should accomplish (e.g., SQL, Prolog).

*OBJECT-ORIENTED: Based on objects that combine data and the functions that operate on that data (e.g., Java, Python).

ASSIGNMENT:

Discuss the following:

1. Low-level Language
2. Assembly Language
- 3 . Domain-specific Language